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Probing the Multi-turn Planning Capabilities of LLMs via 20 Question Games

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Abstract

Large language models (LLMs) are effective at answering questions that are clearly asked. However, when faced with ambiguous queries they can act unpredictably and produce incorrect outputs. This underscores the need for the development of intelligent agents capable of asking clarification questions to resolve ambiguities effectively. This capability requires complex understanding, state tracking, reasoning and planning over multiple conversational turns. However, directly measuring this can be challenging. In this paper, we offer a surrogate problem which assesses an LLMs's capability to deduce an entity unknown to itself, but revealed to a judge, by asking the judge a series of queries. This *entity-deducing game* can serve as an evaluation framework to probe the conversational reasoning and planning capabilities of language models. We systematically evaluate various LLMs and discover significant differences in their performance on this task. We find that strong LLMs like GPT-4 outperform human players by a large margin. We further employ Behavior Cloning (BC) to examine whether a weaker model is capable of imitating a stronger model and generalizing to data or domains, using only the demonstrations from a stronger model. We finally propose to use Reinforcement Learning to enhance reasoning and planning capacity of Vicuna models through episodes of game playing, which lead to significant performance improvement. We hope that this problem offers insights into how autonomous agents could be trained to behave more intelligently in ambiguous circumstances.

1 Introduction

In uncertain circumstances, intelligent conversational agents may need to take the initiative to reduce their uncertainty by asking good questions proactively, thereby solving problems more effectively. This requires intricate, interactive, strategic decision-making and reasoning about the agent's next move in a multi-turn conversation. This capability is crucial in various applications, such as multistep task completion, task-oriented chatbots, recommendations, and conversational search.

Traditionally, dialogue systems, including the clarification process, have been achieved by modularizing various aspects of such tasks into sub-tasks such as *natural language understanding*, *state tracking*, *planning (policy learning)*, and *response generation*. However, recent advances in LLM-powered systems have made it possible to create an end-to-end pipeline, opening up new possibilities for developing autonomous agents that can complete complex tasks using enhanced planning and memory capabilities. Promising works, such as ReAct (Yao et al., 2022), HuggingGPT (Shen et al., 2023), AutoGPT (Significant Gravitas, 2023), LangChain (Langchain-AI, 2023), GPT-Engineer (Anton Osika, 2023) and BabyAGI (Nakajima, 2023), have demonstrated significant potential in this field. These agents require the underlying LLM to retain and recall important information from previous dialogues, resembling the *understanding* and *state tracking* stage. They also rely on the LLM to decompose larger tasks into more manageable components, which is analogous to the *planning*

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stage. Among them, some approaches (*e.g.*, HuggingGPT) use a *static* planning strategy by first generating the complete plan via LLM and subsequently tackling each subtask. Other approaches (*e.g.*, AutoGPT) adopt a *dynamic* and *interactive* planning strategy, where the generation of each action is conditioned on the outcome of the previous planning steps.

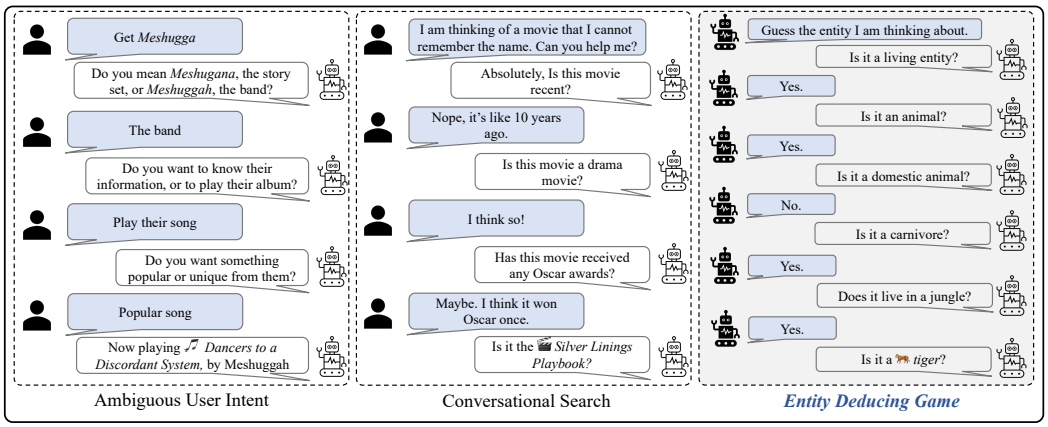

Figure 1: The entity deducing game resembles real scenarios where the agent may need to make strategic decisions regarding the clarification question to be asked based on the current conversation to elicit the actual user intent in as few turns as possible.

While LLM-powered autonomous agents can solve a wide variety of tasks, they can perform poorly when a user's instructions are unclear. This poses a critical challenge – how to accurately elicit and capture the user's intents, which are often nuanced, unpredictable, and noisy, to enable dynamic rather than static human-computer interactions. For example, in Figure 1, the agent has three objectives: 1) accurately assess the current dialog state; 2) eliminate ambiguity in user's intent and satisfy the user demand by asking strategic questions; 3) ask as few questions as possible.

Progress in this direction is difficult because directly measuring complex understanding, reasoning and planning capabilities is challenging. In this study, we investigate this somewhat overlooked research problem – *how good the LLMs are at asking questions and deducing intent.* We propose to use entity-deducing games, specifically the 20 questions game (Q20) (Akinator, 2007), to assess the complex reasoning and strategic planning capability of LLMs in formulating precise questions/guesses over long conversations (Figure 1). This game requires a model to infer an unidentified entity through a sequence of questions that elicit simple responses of "Yes", "No" or "Maybe" with as few queries as possible. To achieve this, the model must be able to track the dialogue state over turns, and use its reasoning and planning skills to effectively partition and narrow down the search scope.

We systematically evaluated several LLMs for their performance on this task, and found significant differences in their conversational reasoning and planning capabilities. We then investigated a set of research questions to enhance open-source models using demonstrations from high-performing closed-source models. Finally, drawing inspiration from Reinforcement Learning from Human Feedback (RLHF) (OpenAI, 2023), we show how PPO (Schulman et al., 2017) can be used to improve vanilla Vicuna (Chiang et al., 2023) models using game playing directly. We provide the code and dataset to facilitate future research [1]. Our findings offer insights into potential future advancements in autonomous agents that proactively lead the conversation.

2 Related Work

Evaluation of Complex Reasoning Capabilities of LLMs There has been extensive work on evaluating the complex reasoning abilities of LLMs(Huang & Chang, 2022). Prior work has created diverse benchmarks, like HELM (Liang et al., 2022) BIG-bench (Srivastava et al., 2022), SuperGLUE (Sarlin et al., 2020), LAMA (Petroni et al., 2019), and CoT-Hub (Fu et al., 2023), which have enabled researchers to assess LLMs across a spectrum of tasks involving arithmetic and math (GSM8K)

[1] `https://github.com/apple/ml-entity-deduction-arena`

(Cobbe et al., 2021), commonsense (ARC) (Clark et al., 2018), symbolic (BBH) (Suzgun et al., 2022), knowledge (MMLU) (Hendrycks et al., 2020), coding (HumanEval) (Chen et al., 2021), and factual (SummEdits) (Laban et al., 2023) have been developed to gauge their real-world abilities. We aim to introduce a novel benchmark to complement existing benchmarks, by evaluating the model's capacity on multi-turn deductive reasoning, state tracking, fact-recalling and strategical planning.

Evaluation of Planning Evaluation of the planning abilities of LLMs is relatively rare. Valmeekam et al. (2022) proposed an assessment framework to gauge the planning capabilities of LLMs to generate valid actions to achieve a special goal and provide a rationale for the plan. Their evaluation on 8 planning tasks reveals LLMs, like GPT-3 (Brown et al., 2020), seem to display a dismal performance. Valmeekam et al. (2023) further evaluates on GPT-4 and suggests the autonomous learning capacity of LLMs to formulate plans is limited and dependent on properly designed heuristics. Xie et al. (2023) similarly indicate that LLMs may encounter difficulties in generating planning goals related to numerical or spatial reasoning, while being responsive to the specific prompts used. Unlike these studies, our task solely relies on textual representations of goals.

Multiturn benchmarks MT-Bench (Zheng et al., 2023) assesses the multi-turn conversation and instruction-following ability of LLMs by annotating their responses to over 80 challenging questions involving writing, role-play, extraction, reasoning, math, coding and knowledge. Bang et al. (2023) evaluates LLMs on 23 different datasets encompassing tasks related to logical reasoning, non-textual reasoning, and commonsense reasoning. The study reveals that incorporating interactive features in LLMs can enhance their performance by employing multi-turn prompt engineering. LMRL Gym (Abdulhai et al., 2023), a concurrent work, investigated Multi-Turn Reinforcement Learning with Language Models across multiple benchmark tasks. In contrast, our work focuses on the depth by presenting a thorough analysis of the entity deduction task.

Entity-deduction game Testing the model's ability to deduct an ambiguous entity or asking clarification questions (Aliannejadi et al., 2019; Cho et al., 2019) has been utilized as a testbed on dialogue systems and visual reasoning tasks. InfoBot (Dhingra et al., 2016) uses reinforcement learning to learn an optimal dialog policy for identifying movie entries from a movie database. ESP (Von Ahn & Dabbish, 2004) and Peekaboom (Von Ahn et al., 2006) demonstrated that deduction games can effectively gather labeled data. GuessWhat?! (De Vries et al., 2017) and ReferIt (Kazemzadeh et al., 2014), assess the visual reasoning capabilities of tested models. These benchmarks involve a model seeing an entity-deduction conversation based on an image to guess the referred object in the image. Our work instead aims to gauge on the model's ability on generating the conversation.

3 Entity-deduction Arena (EDA)

During an entity-deducing game session, two players engage in a collaborative game regarding a specific entity. One player, the "*judge*" (`J`), is provided with the entity and is expected to respond to queries or guesses from the guesser using only the responses "Yes," "No," or "Maybe,". The other player, referred to as the "*guesser*" (`G`) is agnostic about the entity, and is supposed to pose a series of questions to deduce the entity using as few queries as possible.

The judge does not require access to the dialogue history and only needs the entity and current question to provide a response [2]. This task is akin to closed-book QA (Roberts et al., 2020), which state-of-the-art LLMs can reasonably handle. On the other hand, playing the guesser is more demanding. A proficient `G` necessitates several multi-turn dialogue capabilities working in synergy:

1. **State Tracking and Understanding**: `G` must comprehend multi-turn context, track asked questions, and understand its position in the game and coreference resolution.
2. **Strategic Planning**: `G` needs to strategically ask questions to progress efficiently towards a better state, avoiding redundant queries and ensuring consistency with prior knowledge.
3. **Inductive Reasoning**: `G` must use conversation comprehension to generate conjectures based on acquired knowledge. `G` must inherently establish a *taxonomy representation* to efficiently and accurately identify the correct entity among numerous options.

[2] Our experiment indicate that incorporating entire dialogue history negatively impacts accuracy. The additional information tends to confuse the judge rather than improve understanding.